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# This is a R script written in base R (R version 4.3.1).
# Date: August 2023.
# Author: James Wilson.

# For each study and model development, we use the presented regression coefficients
# (log odds ratios) to derive the simplified score weights according to the study's
# methodology

# where the regression coefficients do not correspond to the score weights, differences
# introduced from rounding errors are considered (for example, due to the number of decimal
# places reported in the published tables of regression coefficients)

# many studies reference the method used by Barquet et al for the derivation of risk scores from
# the logistic regression coefficients

## "Points were assigned to each prognostic factor by dividing the coefficient of each predictor from
## the logistic regression analysis by the smallest coefficient, then rounding each quotient to the nearest
## integer. Finally, a prediction score was calculated for each patient by summing the total number of points."

# Barquet N, et al. Prognostic factors in meningococcal disease. Development of a bedside
# predictive model and scoring system. Barcelona Meningococcal Disease Surveillance Group. JAMA.
# 1997 Aug 13;278(6):491-6. doi: 10.1001/jama.278.6.491. PMID: 9256224.

# rc: regression coefficients (log odds ratios)
# aoc: adjusted odds ratios (where regression coefficients not presented)

# rc = natural log (aoc) = ln(aoc)

#####
### de Araujo 2012 ###
#####

rc <- c(
  "other infections" = 1.17,
  "bleeding" = 1.25,
  "jaundice" = 2.31,
  "age_60" = 1.13
)

round(rc / min(rc)) # score is reproducible

# Methodology: Barquet et al. referenced

#####
### Coura-Vital 2014 ###
#####

rc <- c(
  "splenomegaly" = 0.43,
  "edema" = 0.57,
  "weakness" = 0.50,
```

```
"bleeding" = 1.34,  
"jaundice" = 0.48,  
"age_0_5" = 2.14,  
"age_0_5_1" = 1.02,  
"age_19_50" = 0.87,  
"age_50_65" = 1.34,  
"age_65" = 2.25,  
"leishmania-HIV co-infection" = 0.50,  
"bacterial infection" = 0.63  
)  
  
round(rc / min(rc)) # score is reproducible  
  
# methodology: Barquet et al. referenced  
  
#####  
### Werneck 2003 ###  
#####  
  
aor <- c(  
  "diarrhea" = 7.82,  
  "jaundice" = 10.6,  
  "fever" = 8.64,  
  "hematocrit" = 15.9  
)  
  
rc <- log(aor)  
round(rc / min(rc)) # score is reproducible  
  
# methodology: Barquet et al. referenced  
  
#####  
### Sampaio 2010 ###  
#####  
  
aor <- c(  
  "jaundice" = 4.4,  
  "dyspnoea" = 2.8,  
  "mucosal bleeding" = 4.1,  
  "associated infections" = 2.7,  
  "platelets" = 11.7,  
  "neutrophils" = 3.1  
)  
  
rc <- log(aor)  
round(rc / min(rc)) # not reproducible following Barquet's methodology  
round(rc / 0.8) # now it's reproducible - so risk scores are consistent with regression coefficients, but using different  
scaling  
  
# methodology: Barquet et al. referenced
```

```
#####

# correlation with coefficients from Foinquinos 2021
rc <- c(
  "jaundice" = 1.490908,
  "dyspnoea" = 1.015233,
  "mucosal bleeding" = 1.415629,
  "associated infections" = 1.003439, # denominator
  "platelets" = 2.458888,
  "neutrophils" = 1.145403
)

exp(rc) # appear consistent
round(rc / min(rc)) # not reproducible
round(rc / 0.8)

#####
### Foinquinos 2021 ###
#####

rc <- c(
  "jaundice" = 1.490908,
  "dyspnea" = 1.015233,
  "bleeding" = 1.415629,
  "infections" = 1.003439,
  "platelets" = 2.458888,
  "neutrophils" = 1.145403
)

rc_update <- rc * 0.5224832
round((rc_update) / min(rc_update)) # not reproducible

# methodology: Barquet et al (referenced by Sampaio et al)

#####
### Costa 2016 ###
#####

# Regression coefficients reported in Tables 2 and 3
# Risk score coefficients reported in Table 4

# model: < 2 years, clin.
rc <- c(
  "age < 12" = 1.4, # groups switched in article
  "bleeding 3-4 sites" = 1.8,
  "bleeding 5-6 sites" = 3.6,
  "edema" = 1.8,
  "jaundice" = 0.9
)

# included in the risk score, but not presented as regression coefficients: bleeding 1-2 sites, and dysnea coefficients
```

```
rc / min(rc) # not reproducible (score includes dyspnoea)

# model: < 2 years, clin./lab.
rc <- c(
  "age < 12" = 1.1,
  "bleeding 3-4 sites" = 1.8,
  "bleeding 5-6 sites" = 4.1,
  "edema" = 1.4,
  "dyspnea" = 1.1,
  "AST/ALT" = 2.5
)
# included in the risk score, but not presented as regression coefficients: bleeding 1-2 sites

rc / min(rc) # not reproducible
rc / 0.95 # changing the denominator does not reproduce scores

2.55 / 1.05 # max possible score for AST/ALT with rounding error

# model: > or equal to 2 years, clin.
aor <- c(
  "age >40" = 3.5,
  "2 sites(?exp)" = 2.3, # inconsistent with rc, exp(2.3)
  "3 or more sites" = 10.4, # inconsistent with rc, exp(2.6)
  "HIV" = 6.1,
  "edema" = 2.2,
  "jaundice" = 2.9,
  "dyspnea" = 2.3,
  "vomiting" = 2.0,
  "bac infection" = 2.9
)

rc <- c(
  "age >40" = 1.3,
  "2 sites(?exp)" = 2.3, # inconsistent with rc, log(2.3)
  "3 or more sites" = 2.6, # inconsistent with rc, log(10.4)
  "HIV" = 1.8,
  "edema" = 0.8,
  "jaundice" = 1.1,
  "dyspnea" = 0.8,
  "vomiting" = 0.7,
  "bac infection" = 1.1
)
# included in the risk score, but not presented as regression coefficients: age 15-40 years, bleeding categories differ

log(aor)
log(aor) / min(log(aor))
round(log(aor) / min(log(aor))) # not reproducible (different grouping)

# model: > or equal to 2 years, clin./lab.
rc <- c(
```

```
"3 or more sites" = 1.7, # not significant predictor
"hiv" = 2.2,
"jaundice" = 1.0,
"dyspnea" = 0.8,
"bac infection" = 1.2,
"leuk" = 1.3,
"plat" = 2.0,
"renal fail" = 2.4
)
# "3 or more sites" does not appear in risk score

rc / min(rc)
round(rc / min(rc)) # not reproducible

# methodology: Barquet et al. referenced

#####
### Abongomera 2017 ###
#####

alhr <- c(
  ">40" = 2.22,
  "5-18" = 0.93,
  "HIV pos" = 3.04,
  "HIV neg" = 0.54,
  "Hb < 6.5" = 2.16,
  "Hb > 6.5" = 0.67,
  "Bleeding +" = 3.11,
  "Bleeding -" = 0.93,
  "Jaundice +" = 3.21,
  "Jaundice -" = 0.93,
  "Weakness +" = 1.64,
  "Weakness -" = 0.84,
  "Edema -" = 0.84,
  "Edema +" = 2.36,
  "Ascites +" = 5.84,
  "Ascites -" = 0.92,
  "TB +" = 1.71,
  "TB -" = 0.92
)

alhr_adj <- alhr[alhr >= 1.5 | alhr < 0.67]
round(log(alhr_adj)) # reproducible

# methodology: Speigalhalter-Knill-Jones (Barquet not referenced)

#####
### Kamink 2017 ###
#####

rc <- c(
```

```
"age_2_5" = 0.98,  
"age_2" = 2.37,  
"weak_sev" = 0.92,  
"weak_col" = 3.13,  
"hb_75_89" = 0.57,  
"hb_60_74" = 1.02,  
"hb_60" = 2.05,  
"jaundice" = 2.55  
)  
  
round(rc / min(rc)) # reproducible  
  
rc <- c(  
  "weak_sev" = 0.90,  
  "weak_col" = 1.44,  
  "bmi_145_159" = 0.96,  
  "bmi_130_144" = 1.02,  
  "bmi_13" = 1.58,  
  # "hb_75_89" = 0.07,  
  # "hb_60_74" = 0.42,  
  "hb_60" = 1.50,  
  "oed_asc" = 1.58,  
  "jaundice" = 1.23  
)  
  
round(rc / min(rc)) # reproducible  
  
# methodology: Barquet et al. referenced
```